Lightgcn baseline smoke test 數據

PS D:\Research\SCA-Recommendation> py run.py --config configs/lightgcn.yaml [INFO] Using device: cpu [INFO] Model type: lightgcn [INFO] Data loaded | num_users=6041, num_items=3953, train_pairs=988129 [DIAG] ===== Data Split Inspection ===== [DIAG] num_users=6041, num_items=3953 [DIAG] train_pairs=988129 [DIAG] valid_pairs=6040 [DIAG] test_pairs=6040 [DIAG] train_users=6040 [DIAG] valid_users=6040 [DIAG] test_users=6040 [DIAG] train-valid overlap users=0, items=0 [DIAG] train-test overlap users=0, items=0 [DIAG] valid-test overlap users=0, items=0 [DIAG] valid gt per user | min=1, max=1, avg=1.0000 [DIAG] test gt per user | min=1, max=1, avg=1.0000 [DIAG] user=1 | train=51 | valid=1 | test=1 | candidate=3901 [DIAG] user=2 | train=127 | valid=1 | test=1 | candidate=3825 [DIAG] user=3 | train=49 | valid=1 | test=1 | candidate=3903 [DIAG] user=4 | train=19 | valid=1 | test=1 | candidate=3933 [DIAG] user=5 | train=196 | valid=1 | test=1 | candidate=3756 [DIAG] user=6 | train=69 | valid=1 | test=1 | candidate=3883 [DIAG] user=7 | train=29 | valid=1 | test=1 | candidate=3923 [DIAG] user=8 | train=137 | valid=1 | test=1 | candidate=3815 [DIAG] user=9 | train=104 | valid=1 | test=1 | candidate=3848 [DIAG] user=10 | train=399 | valid=1 | test=1 | candidate=3553 [DIAG] user=11 | train=135 | valid=1 | test=1 | candidate=3817 [DIAG] user=12 | train=21 | valid=1 | test=1 | candidate=3931 [DIAG] user=13 | train=106 | valid=1 | test=1 | candidate=3846 [DIAG] user=14 | train=23 | valid=1 | test=1 | candidate=3929 [DIAG] user=15 | train=199 | valid=1 | test=1 | candidate=3753 [DIAG] user=16 | train=33 | valid=1 | test=1 | candidate=3919 [DIAG] user=17 | train=209 | valid=1 | test=1 | candidate=3743 [DIAG] user=18 | train=303 | valid=1 | test=1 | candidate=3649 [DIAG] user=19 | train=253 | valid=1 | test=1 | candidate=3699 [DIAG] user=20 | train=22 | valid=1 | test=1 | candidate=3930 [DIAG] sampled candidate size | min=3553, max=3933, avg=3827.80 [DIAG] ===== End Inspection ===== [INFO] Batch inspection: {'batch_size': 1024, 'user_ids_shape': (1024,), 'pos_item_ids_shape': (1024,), 'neg_item_ids_shape': (1024,), 'user_all_embeddings_shape': (6041, 64), 'item_all_embeddings_shape': (3953, 64), 'user_emb_shape': (1024, 64), 'pos_emb_shape': (1024, 64), 'neg_emb_shape': (1024, 64), 'bpr_loss': 0.6931420564651489, 'pos_mean': 1.6686613889760338e-05, 'neg_mean': 6.343602763081435e-06, 'pos_gt_neg': 0.517578125} [Epoch 001] loss=0.692471 | bpr=0.692467 | reg=0.045280 | pos_mean=0.002249 | neg_mean=0.000887 | pos>neg=0.759814 [Epoch 002] loss=0.683050 | bpr=0.683042 | reg=0.076431 | pos_mean=0.034091 | neg_mean=0.013613 | pos>neg=0.863574 [INFO] Training finished. [INFO] Start evaluation... [RESULT] Test Metrics: {'Recall@10': 0.03609271523178808, 'NDCG@10': 0.017766570566454976, 'HR@10': 0.03609271523178808}

改進優化加速後版本

Scaoff

PS D:\Research\SCA-Recommendation> py run.py --config configs/sca_off.yaml
[INFO] Using device: cpu
[INFO] Model type: sca
[INFO] Data loaded | num_users=6041, num_items=3953, train_pairs=988129
[DIAG] ===== Data Split Inspection =====
[DIAG] num_users=6041, num_items=3953

```
[DIAG] train_pairs=988129
[DIAG] valid_pairs=6040
[DIAG] test_pairs=6040
[DIAG] train_users=6040
[DIAG] valid_users=6040
[DIAG] test_users=6040
[DIAG] train-valid overlap users=0, items=0
[DIAG] train-test overlap users=0, items=0
[DIAG] valid-test overlap users=0, items=0
[DIAG] valid gt per user | min=1, max=1, avg=1.0000
[DIAG] test gt per user | min=1, max=1, avg=1.0000
[DIAG] user=1 | train=51 | valid=1 | test=1 | candidate=3901
[DIAG] user=2 | train=127 | valid=1 | test=1 | candidate=3825
[DIAG] user=3 | train=49 | valid=1 | test=1 | candidate=3903
[DIAG] user=4 | train=19 | valid=1 | test=1 | candidate=3933
[DIAG] user=5 | train=196 | valid=1 | test=1 | candidate=3756
[DIAG] user=6 | train=69 | valid=1 | test=1 | candidate=3883
[DIAG] user=7 | train=29 | valid=1 | test=1 | candidate=3923
[DIAG] user=8 | train=137 | valid=1 | test=1 | candidate=3815
[DIAG] user=9 | train=104 | valid=1 | test=1 | candidate=3848
[DIAG] user=10 | train=399 | valid=1 | test=1 | candidate=3553
[DIAG] user=11 | train=135 | valid=1 | test=1 | candidate=3817
[DIAG] user=12 | train=21 | valid=1 | test=1 | candidate=3931
[DIAG] user=13 | train=106 | valid=1 | test=1 | candidate=3846
[DIAG] user=14 | train=23 | valid=1 | test=1 | candidate=3929
[DIAG] user=15 | train=199 | valid=1 | test=1 | candidate=3753
[DIAG] user=16 | train=33 | valid=1 | test=1 | candidate=3919
[DIAG] user=17 | train=209 | valid=1 | test=1 | candidate=3743
[DIAG] user=18 | train=303 | valid=1 | test=1 | candidate=3649
[DIAG] user=19 | train=253 | valid=1 | test=1 | candidate=3699
[DIAG] user=20 | train=22 | valid=1 | test=1 | candidate=3930
[DIAG] sampled candidate size | min=3553, max=3933, avg=3827.80
[DIAG] ===== End Inspection =====
[INFO] Batch inspection: {'user_ids': (4096,), 'pos_item_ids': (4096,), 'neg_item_ids': (4096,),
'user_emb': (4096, 64), 'pos_item_emb': (4096, 64), 'neg_item_emb': (4096, 64), 'e_u': (4096,
64), 'c_u': (4096, 64), 'z_u': (4096, 32), 'delta_u': (4096, 64), 'g_u': (4096, 64), 'pos_scores':
(4096,), 'neg_scores': (4096,)}
[Epoch 001] loss=0.804428 | bpr=0.693138 | align=1.090836 | reg=22.057799 |
pos_mean=0.000016 | neg_mean=-0.000002 | pos>neg=0.527995 | delta_abs=0.113376 |
gate_mean=0.502251 | gate_std=0.025405 | ctrl_shift=0.000000
[Epoch 002] loss=0.719964 | bpr=0.693118 | align=0.245290 | reg=23.169389 |
pos_mean=0.000102 | neg_mean=0.000045 | pos>neg=0.577230 | delta_abs=0.112907 |
gate_mean=0.502348 | gate_std=0.025523 | ctrl_shift=0.000000
[INFO] Training finished.
```

```
[INFO] Start evaluation...
[RESULT] Test Metrics:
{'Recall@10': 0.008112582781456953, 'NDCG@10': 0.004045497470189505, 'HR@10':
0.008112582781456953}
```

Scion

```
PS D:\Research\SCA-Recommendation> py run.py --config configs/sca_on.yaml
[INFO] Using device: cpu
[INFO] Model type: sca
[INFO] Data loaded | num_users=6041, num_items=3953, train_pairs=988129
[DIAG] ===== Data Split Inspection =====
[DIAG] num_users=6041, num_items=3953
[DIAG] train_pairs=988129
[DIAG] valid_pairs=6040
[DIAG] test_pairs=6040
[DIAG] train_users=6040
[DIAG] valid_users=6040
[DIAG] test_users=6040
[DIAG] train-valid overlap users=0, items=0
[DIAG] train-test overlap users=0, items=0
[DIAG] valid-test overlap users=0, items=0
[DIAG] valid gt per user | min=1, max=1, avg=1.0000
[DIAG] test gt per user | min=1, max=1, avg=1.0000
[DIAG] user=1 | train=51 | valid=1 | test=1 | candidate=3901
[DIAG] user=2 | train=127 | valid=1 | test=1 | candidate=3825
[DIAG] user=3 | train=49 | valid=1 | test=1 | candidate=3903
[DIAG] user=4 | train=19 | valid=1 | test=1 | candidate=3933
[DIAG] user=5 | train=196 | valid=1 | test=1 | candidate=3756
[DIAG] user=6 | train=69 | valid=1 | test=1 | candidate=3883
[DIAG] user=7 | train=29 | valid=1 | test=1 | candidate=3923
[DIAG] user=8 | train=137 | valid=1 | test=1 | candidate=3815
[DIAG] user=9 | train=104 | valid=1 | test=1 | candidate=3848
[DIAG] user=10 | train=399 | valid=1 | test=1 | candidate=3553
[DIAG] user=11 | train=135 | valid=1 | test=1 | candidate=3817
[DIAG] user=12 | train=21 | valid=1 | test=1 | candidate=3931
[DIAG] user=13 | train=106 | valid=1 | test=1 | candidate=3846
[DIAG] user=14 | train=23 | valid=1 | test=1 | candidate=3929
[DIAG] user=15 | train=199 | valid=1 | test=1 | candidate=3753
[DIAG] user=16 | train=33 | valid=1 | test=1 | candidate=3919
[DIAG] user=17 | train=209 | valid=1 | test=1 | candidate=3743
[DIAG] user=18 | train=303 | valid=1 | test=1 | candidate=3649
[DIAG] user=19 | train=253 | valid=1 | test=1 | candidate=3699
[DIAG] user=20 | train=22 | valid=1 | test=1 | candidate=3930
[DIAG] sampled candidate size | min=3553, max=3933, avg=3827.80
```

[DIAG] ===== End Inspection =====
[INFO] Batch inspection: {'user_ids': (4096,), 'pos_item_ids': (4096,), 'neg_item_ids': (4096,), 'user_emb': (4096, 64), 'pos_item_emb': (4096, 64), 'neg_item_emb': (4096, 64), 'e_u': (4096, 64), 'c_u': (4096, 64), 'z_u': (4096, 32), 'delta_u': (4096, 64), 'g_u': (4096, 64), 'pos_scores': (4096,), 'neg_scores': (4096,)}
[Epoch 001] loss=0.929646 | bpr=0.693218 | align=1.090836 | reg=1273.442586 | pos_mean=-0.000179 | neg_mean=-0.000043 | pos>neg=0.485677 | delta_abs=0.113376 | gate_mean=0.502251 | gate_std=0.025405 | ctrl_shift=0.056929
[Epoch 002] loss=0.837057 | bpr=0.691808 | align=0.259674 | reg=1192.812419 | pos_mean=0.005205 | neg_mean=0.002519 | pos>neg=0.706217 | delta_abs=0.108022 | gate_mean=0.500148 | gate_std=0.024498 | ctrl_shift=0.054007
[INFO] Training finished.
[INFO] Start evaluation...
[RESULT] Test Metrics:
{'Recall@10': 0.02847682119205298, 'NDCG@10': 0.013708923861193512, 'HR@10': 0.02847682119205298}

雖然都是smoke test版本，但不足以基於同一公平性原則來相互比較

第一阶段的“最小实验闭环验证”已经跑通了，但还不是正式 **Table 1**。
你现在已经证明了三件事：

LightGCN 能正常训练、评估、出指标。
SCA-off 能正常训练，而且 `ctrl_shift=0`，说明 control 确实被关掉了。
SCA-on 能正常训练，而且 `ctrl_shift≈0.05`，说明 semantic control 确实在起作用。

这已经是很重要的阶段性成果。

现在把三组 smoke test 放一起看：

Model	Recall@10	NDCG@10	HR@10	机制信号
LightGCN-smoke	0.0361	0.0178	0.0361	baseline
SCA-off-smoke	0.0081	0.0040	0.0081	`ctrl_shift=0`
SCA-on-smoke	0.0285	0.0137	0.0285	`ctrl_shift>0`

这张表不能放论文，但它很有价值，因为它回答了你现在最关键的问题：项目是不是正常的。

答案是：正常，而且机制链路成立。

为什么我这么判断：

第一，`SCA-on` 比 `SCA-off` 明显高。
这是最关键的现象。虽然现在还不是正式训练，但在同样的 smoke test 条件下，`SCA-on` 从 `0.0081` 提到 `0.0285`，差距很明显。这说明 semantic control 不是摆设。

第二，`ctrl_shift` 的行为完全符合预期。
`SCA-off` 里是 `0.000000`，`SCA-on` 里是 `0.056929 -> 0.054007`。这几乎就是你论文里最想要的“机制证据”：off 时不动，on 时真动。

第三，`gate_mean` 和 `gate_std` 也正常。
大概在 `0.50` 左右，std 也不是接近 0，说明 gate 没塌缩成恒定开关。

第四，`SCA-on` 的训练趋势也正常。
`bpr` 从 `0.693218` 到 `0.691808`，`pos_mean` 高于 `neg_mean`，`pos>neg` 从 `0.485677` 到 `0.706217`，说明它在学习，不是乱跑。

当然，也有一个你必须正视的现实：

当前 **SCA-off** 明显低于 **LightGCN**，不是一个好现象。
但我现在不把它解读成方法有问题，而是更大概率说明：
你目前这个加速版 `trainer_sca.py` 作为 smoke test 可以验证机制，但还不是和 LightGCN 完全公平、完全成熟的正式训练实现。

换句话说，现在这三组结果能支持：

- 代码没坏
- evaluator 没坏
- SCA 机制在工作
- off/on 有区分度

但还不能支持：

- “SCA 正式优于 LightGCN”
- “论文主结果成立”

所以你现在不是失败了，而是已经完成了第一阶段里的正确性验证，接下来进入正式实验准备阶段。

中間版本 baseline off on
Lightgcn
PS D:\Research\SCA-Recommendation> py run.py --config configs/lightgcn_mid.yaml
[INFO] Using device: cpu
[INFO] Model type: lightgcn
[INFO] Data loaded | num_users=6041, num_items=3953, train_pairs=988129
[DIAG] ===== Data Split Inspection =====
[DIAG] num_users=6041, num_items=3953
[DIAG] train_pairs=988129
[DIAG] valid_pairs=6040
[DIAG] test_pairs=6040
[DIAG] train_users=6040
[DIAG] valid_users=6040
[DIAG] test_users=6040
[DIAG] train-valid overlap users=0, items=0
[DIAG] train-test overlap users=0, items=0
[DIAG] valid-test overlap users=0, items=0
[DIAG] valid gt per user | min=1, max=1, avg=1.0000
[DIAG] test gt per user | min=1, max=1, avg=1.0000
[DIAG] user=1 | train=51 | valid=1 | test=1 | candidate=3901
[DIAG] user=2 | train=127 | valid=1 | test=1 | candidate=3825
[DIAG] user=3 | train=49 | valid=1 | test=1 | candidate=3903
[DIAG] user=4 | train=19 | valid=1 | test=1 | candidate=3933
[DIAG] user=5 | train=196 | valid=1 | test=1 | candidate=3756
[DIAG] user=6 | train=69 | valid=1 | test=1 | candidate=3883
[DIAG] user=7 | train=29 | valid=1 | test=1 | candidate=3923
[DIAG] user=8 | train=137 | valid=1 | test=1 | candidate=3815
[DIAG] user=9 | train=104 | valid=1 | test=1 | candidate=3848
[DIAG] user=10 | train=399 | valid=1 | test=1 | candidate=3553
[DIAG] user=11 | train=135 | valid=1 | test=1 | candidate=3817
[DIAG] user=12 | train=21 | valid=1 | test=1 | candidate=3931
[DIAG] user=13 | train=106 | valid=1 | test=1 | candidate=3846
[DIAG] user=14 | train=23 | valid=1 | test=1 | candidate=3929
[DIAG] user=15 | train=199 | valid=1 | test=1 | candidate=3753
[DIAG] user=16 | train=33 | valid=1 | test=1 | candidate=3919
[DIAG] user=17 | train=209 | valid=1 | test=1 | candidate=3743
[DIAG] user=18 | train=303 | valid=1 | test=1 | candidate=3649
[DIAG] user=19 | train=253 | valid=1 | test=1 | candidate=3699
[DIAG] user=20 | train=22 | valid=1 | test=1 | candidate=3930
[DIAG] sampled candidate size | min=3553, max=3933, avg=3827.80
[DIAG] ===== End Inspection =====
[INFO] Batch inspection: {'batch_size': 4096, 'user_ids_shape': (4096,), 'pos_item_ids_shape': (4096,), 'neg_item_ids_shape': (4096,), 'user_all_embeddings_shape': (6041, 64),

'item_all_embeddings_shape': (3953, 64), 'user_emb_shape': (4096, 64), 'pos_emb_shape': (4096, 64), 'neg_emb_shape': (4096, 64), 'bpr_loss': 0.6931405067443848, 'pos_mean': 1.2868271369370632e-05, 'neg_mean': -4.4368965745889e-07, 'pos_gt_neg': 0.518798828125}
[Epoch 001] loss=0.678958 | bpr=0.678949 | reg=0.091112 | pos_mean=0.048324 | neg_mean=0.018953 | pos>neg=0.818984
[Epoch 002] loss=0.548003 | bpr=0.547940 | reg=0.626296 | pos_mean=0.594179 | neg_mean=0.229754 | pos>neg=0.863086
[Epoch 003] loss=0.400277 | bpr=0.400111 | reg=1.654496 | pos_mean=1.646545 | neg_mean=0.631123 | pos>neg=0.863667
[Epoch 004] loss=0.345506 | bpr=0.345255 | reg=2.516961 | pos_mean=2.526247 | neg_mean=0.954233 | pos>neg=0.864375
[Epoch 005] loss=0.331524 | bpr=0.331221 | reg=3.039636 | pos_mean=3.055429 | neg_mean=1.141823 | pos>neg=0.863618
[Epoch 006] loss=0.325245 | bpr=0.324911 | reg=3.345994 | pos_mean=3.351239 | neg_mean=1.232261 | pos>neg=0.864438
[Epoch 007] loss=0.324177 | bpr=0.323826 | reg=3.513734 | pos_mean=3.501188 | neg_mean=1.272226 | pos>neg=0.863911
[Epoch 008] loss=0.322922 | bpr=0.322561 | reg=3.617976 | pos_mean=3.572207 | neg_mean=1.279764 | pos>neg=0.864512
[Epoch 009] loss=0.323007 | bpr=0.322638 | reg=3.682168 | pos_mean=3.596206 | neg_mean=1.279446 | pos>neg=0.863086
[Epoch 010] loss=0.321446 | bpr=0.321073 | reg=3.728300 | pos_mean=3.608158 | neg_mean=1.265933 | pos>neg=0.864399
[INFO] Training finished.
[INFO] Start evaluation...
[RESULT] Test Metrics:
{'Recall@10': 0.03559602649006623, 'NDCG@10': 0.017518074623627727, 'HR@10': 0.03559602649006623}

Off
PS D:\Research\SCA-Recommendation> py run.py --config configs/sca_off_mid.yaml
[INFO] Using device: cpu
[INFO] Model type: sca
[INFO] Data loaded | num_users=6041, num_items=3953, train_pairs=988129
[DIAG] ===== Data Split Inspection =====
[DIAG] num_users=6041, num_items=3953
[DIAG] train_pairs=988129
[DIAG] valid_pairs=6040
[DIAG] test_pairs=6040
[DIAG] train_users=6040
[DIAG] valid_users=6040
[DIAG] test_users=6040

```
[DIAG] train-valid overlap users=0, items=0
[DIAG] train-test overlap users=0, items=0
[DIAG] valid-test overlap users=0, items=0
[DIAG] valid gt per user | min=1, max=1, avg=1.0000
[DIAG] test gt per user | min=1, max=1, avg=1.0000
[DIAG] user=1 | train=51 | valid=1 | test=1 | candidate=3901
[DIAG] user=2 | train=127 | valid=1 | test=1 | candidate=3825
[DIAG] user=3 | train=49 | valid=1 | test=1 | candidate=3903
[DIAG] user=4 | train=19 | valid=1 | test=1 | candidate=3933
[DIAG] user=5 | train=196 | valid=1 | test=1 | candidate=3756
[DIAG] user=6 | train=69 | valid=1 | test=1 | candidate=3883
[DIAG] user=7 | train=29 | valid=1 | test=1 | candidate=3923
[DIAG] user=8 | train=137 | valid=1 | test=1 | candidate=3815
[DIAG] user=9 | train=104 | valid=1 | test=1 | candidate=3848
[DIAG] user=10 | train=399 | valid=1 | test=1 | candidate=3553
[DIAG] user=11 | train=135 | valid=1 | test=1 | candidate=3817
[DIAG] user=12 | train=21 | valid=1 | test=1 | candidate=3931
[DIAG] user=13 | train=106 | valid=1 | test=1 | candidate=3846
[DIAG] user=14 | train=23 | valid=1 | test=1 | candidate=3929
[DIAG] user=15 | train=199 | valid=1 | test=1 | candidate=3753
[DIAG] user=16 | train=33 | valid=1 | test=1 | candidate=3919
[DIAG] user=17 | train=209 | valid=1 | test=1 | candidate=3743
[DIAG] user=18 | train=303 | valid=1 | test=1 | candidate=3649
[DIAG] user=19 | train=253 | valid=1 | test=1 | candidate=3699
[DIAG] user=20 | train=22 | valid=1 | test=1 | candidate=3930
[DIAG] sampled candidate size | min=3553, max=3933, avg=3827.80
[DIAG] ===== End Inspection =====
[INFO] Batch inspection: {'user_ids': (4096,), 'pos_item_ids': (4096,), 'neg_item_ids': (4096,),
'user_emb': (4096, 64), 'pos_item_emb': (4096, 64), 'neg_item_emb': (4096, 64), 'e_u': (4096,
64), 'c_u': (4096, 64), 'z_u': (4096, 32), 'delta_u': (4096, 64), 'g_u': (4096, 64), 'pos_scores':
(4096,), 'neg_scores': (4096,)}
[Epoch 001] loss=0.800744 | bpr=0.693140 | align=1.053983 | reg=22.059677 |
pos_mean=0.000014 | neg_mean=0.000001 | pos>neg=0.519536 | delta_abs=0.113387 |
gate_mean=0.502258 | gate_std=0.025406 | ctrl_shift=0.000000
[Epoch 002] loss=0.719187 | bpr=0.693117 | align=0.237519 | reg=23.181310 |
pos_mean=0.000109 | neg_mean=0.000048 | pos>neg=0.579214 | delta_abs=0.112752 |
gate_mean=0.502235 | gate_std=0.025569 | ctrl_shift=0.000000
[Epoch 003] loss=0.706156 | bpr=0.693087 | align=0.106025 | reg=24.669055 |
pos_mean=0.000232 | neg_mean=0.000111 | pos>neg=0.640103 | delta_abs=0.114262 |
gate_mean=0.502203 | gate_std=0.025784 | ctrl_shift=0.000000
[Epoch 004] loss=0.702147 | bpr=0.693052 | align=0.064606 | reg=26.347780 |
pos_mean=0.000371 | neg_mean=0.000181 | pos>neg=0.694248 | delta_abs=0.116344 |
gate_mean=0.502129 | gate_std=0.026023 | ctrl_shift=0.000000
```

```
[Epoch 005] loss=0.700681 | bpr=0.693015 | align=0.048588 | reg=28.067816 |
pos_mean=0.000514 | neg_mean=0.000250 | pos>neg=0.734800 | delta_abs=0.118603 |
gate_mean=0.502028 | gate_std=0.026279 | ctrl_shift=0.000000
[Epoch 006] loss=0.699982 | bpr=0.692981 | align=0.040223 | reg=29.786984 |
pos_mean=0.000655 | neg_mean=0.000323 | pos>neg=0.761450 | delta_abs=0.120835 |
gate_mean=0.501909 | gate_std=0.026546 | ctrl_shift=0.000000
[Epoch 007] loss=0.699502 | bpr=0.692948 | align=0.034130 | reg=31.409182 |
pos_mean=0.000790 | neg_mean=0.000390 | pos>neg=0.779790 | delta_abs=0.122940 |
gate_mean=0.501782 | gate_std=0.026816 | ctrl_shift=0.000000
[Epoch 008] loss=0.699087 | bpr=0.692915 | align=0.028799 | reg=32.922241 |
pos_mean=0.000917 | neg_mean=0.000452 | pos>neg=0.794761 | delta_abs=0.124884 |
gate_mean=0.501657 | gate_std=0.027084 | ctrl_shift=0.000000
[Epoch 009] loss=0.698715 | bpr=0.692885 | align=0.023947 | reg=34.353050 |
pos_mean=0.001036 | neg_mean=0.000511 | pos>neg=0.804561 | delta_abs=0.126667 |
gate_mean=0.501540 | gate_std=0.027348 | ctrl_shift=0.000000
[Epoch 010] loss=0.698390 | bpr=0.692857 | align=0.019649 | reg=35.686286 |
pos_mean=0.001147 | neg_mean=0.000566 | pos>neg=0.811445 | delta_abs=0.128296 |
gate_mean=0.501434 | gate_std=0.027605 | ctrl_shift=0.000000
[INFO] Training finished.
[INFO] Start evaluation...
[RESULT] Test Metrics:
{'Recall@10': 0.031788079470198675, 'NDCG@10': 0.014757334191499587, 'HR@10':
0.031788079470198675}
```

On

```
PS D:\Research\SCA-Recommendation> py run.py --config configs/sca_on_mid.yaml
[INFO] Using device: cpu
[INFO] Model type: sca
[INFO] Data loaded | num_users=6041, num_items=3953, train_pairs=988129
[DIAG] ===== Data Split Inspection =====
[DIAG] num_users=6041, num_items=3953
[DIAG] train_pairs=988129
[DIAG] valid_pairs=6040
[DIAG] test_pairs=6040
[DIAG] train_users=6040
[DIAG] valid_users=6040
[DIAG] test_users=6040
[DIAG] train-valid overlap users=0, items=0
[DIAG] train-test overlap users=0, items=0
[DIAG] valid-test overlap users=0, items=0
[DIAG] valid gt per user | min=1, max=1, avg=1.0000
[DIAG] test gt per user | min=1, max=1, avg=1.0000
[DIAG] user=1 | train=51 | valid=1 | test=1 | candidate=3901
```

```
[DIAG] user=2 | train=127 | valid=1 | test=1 | candidate=3825
[DIAG] user=3 | train=49 | valid=1 | test=1 | candidate=3903
[DIAG] user=4 | train=19 | valid=1 | test=1 | candidate=3933
[DIAG] user=5 | train=196 | valid=1 | test=1 | candidate=3756
[DIAG] user=6 | train=69 | valid=1 | test=1 | candidate=3883
[DIAG] user=7 | train=29 | valid=1 | test=1 | candidate=3923
[DIAG] user=8 | train=137 | valid=1 | test=1 | candidate=3815
[DIAG] user=9 | train=104 | valid=1 | test=1 | candidate=3848
[DIAG] user=10 | train=399 | valid=1 | test=1 | candidate=3553
[DIAG] user=11 | train=135 | valid=1 | test=1 | candidate=3817
[DIAG] user=12 | train=21 | valid=1 | test=1 | candidate=3931
[DIAG] user=13 | train=106 | valid=1 | test=1 | candidate=3846
[DIAG] user=14 | train=23 | valid=1 | test=1 | candidate=3929
[DIAG] user=15 | train=199 | valid=1 | test=1 | candidate=3753
[DIAG] user=16 | train=33 | valid=1 | test=1 | candidate=3919
[DIAG] user=17 | train=209 | valid=1 | test=1 | candidate=3743
[DIAG] user=18 | train=303 | valid=1 | test=1 | candidate=3649
[DIAG] user=19 | train=253 | valid=1 | test=1 | candidate=3699
[DIAG] user=20 | train=22 | valid=1 | test=1 | candidate=3930
[DIAG] sampled candidate size | min=3553, max=3933, avg=3827.80
[DIAG] ===== End Inspection =====
[INFO] Batch inspection: {'user_ids': (4096,), 'pos_item_ids': (4096,), 'neg_item_ids': (4096,),
'user_emb': (4096, 64), 'pos_item_emb': (4096, 64), 'neg_item_emb': (4096, 64), 'e_u': (4096,
64), 'c_u': (4096, 64), 'z_u': (4096, 32), 'delta_u': (4096, 64), 'g_u': (4096, 64), 'pos_scores':
(4096,), 'neg_scores': (4096,)}
[Epoch 001] loss=0.925896 | bpr=0.693163 | align=1.053983 | reg=1273.351721 |
pos_mean=-0.000077 | neg_mean=-0.000051 | pos>neg=0.496313 | delta_abs=0.113387 |
gate_mean=0.502258 | gate_std=0.025406 | ctrl_shift=0.056938
[Epoch 002] loss=0.836627 | bpr=0.691813 | align=0.250429 | reg=1197.712659 |
pos_mean=0.005129 | neg_mean=0.002452 | pos>neg=0.702139 | delta_abs=0.108055 |
gate_mean=0.500221 | gate_std=0.024502 | ctrl_shift=0.054036
[Epoch 003] loss=0.815892 | bpr=0.690950 | align=0.128724 | reg=1120.698181 |
pos_mean=0.008440 | neg_mean=0.004032 | pos>neg=0.772036 | delta_abs=0.103418 |
gate_mean=0.498326 | gate_std=0.023650 | ctrl_shift=0.051510
[Epoch 004] loss=0.804263 | bpr=0.690365 | align=0.094785 | reg=1044.188462 |
pos_mean=0.010664 | neg_mean=0.005081 | pos>neg=0.798311 | delta_abs=0.098984 |
gate_mean=0.496547 | gate_std=0.022791 | ctrl_shift=0.049114
[Epoch 005] loss=0.795295 | bpr=0.689945 | align=0.084185 | reg=969.317201 |
pos_mean=0.012191 | neg_mean=0.005763 | pos>neg=0.813213 | delta_abs=0.094487 |
gate_mean=0.494894 | gate_std=0.021911 | ctrl_shift=0.046716
[Epoch 006] loss=0.787290 | bpr=0.689640 | align=0.079637 | reg=896.859128 |
pos_mean=0.013241 | neg_mean=0.006200 | pos>neg=0.823882 | delta_abs=0.089838 |
gate_mean=0.493399 | gate_std=0.021009 | ctrl_shift=0.044275
```

[Epoch 007] loss=0.779806 | bpr=0.689437 | align=0.076196 | reg=827.498301 | pos_mean=0.013945 | neg_mean=0.006495 | pos>neg=0.829849 | delta_abs=0.085093 | gate_mean=0.492080 | gate_std=0.020095 | ctrl_shift=0.041818
[Epoch 008] loss=0.772750 | bpr=0.689295 | align=0.072788 | reg=761.752968 | pos_mean=0.014386 | neg_mean=0.006651 | pos>neg=0.834312 | delta_abs=0.080352 | gate_mean=0.490945 | gate_std=0.019177 | ctrl_shift=0.039392
[Epoch 009] loss=0.766150 | bpr=0.689204 | align=0.069513 | reg=699.949248 | pos_mean=0.014618 | neg_mean=0.006698 | pos>neg=0.838423 | delta_abs=0.075705 | gate_mean=0.489997 | gate_std=0.018265 | ctrl_shift=0.037037
[Epoch 010] loss=0.760077 | bpr=0.689176 | align=0.066747 | reg=642.261229 | pos_mean=0.014680 | neg_mean=0.006706 | pos>neg=0.840825 | delta_abs=0.071233 | gate_mean=0.489228 | gate_std=0.017367 | ctrl_shift=0.034791
[INFO] Training finished.
[INFO] Start evaluation...
[RESULT] Test Metrics:
{'Recall@10': 0.037251655629139076, 'NDCG@10': 0.01776007550556556, 'HR@10': 0.037251655629139076}

现在的中间版结果

Model	Recall@10	NDCG@10	HR@10
LightGCN-mid	0.0356	0.0175	0.0356
SCA-off-mid	0.0318	0.0148	0.0318
SCA-on-mid	0.0373	0.0178	0.0373

我对这组结果的判断

这组结果已经比 smoke test 有意义得多了，而且趋势是对的。

第一，**SCA-on > SCA-off**，而且差距是清楚的。
这说明 semantic control 打开之后，模型确实带来了增益，不是摆设。

第二，**SCA-on** 略高于 **LightGCN**。
提升不大，但方向是对的，而且这正是现在这个阶段最重要的事：你的主张开始有实验支撑了。

第三，**SCA-off** 低于 **SCA-on**，也低于 **LightGCN**。
这也合理，因为它保留了 SCA 的框架成本，但把真正有效的 control 关掉了。

第四，机制日志非常关键，而且也支持论文叙事：

- `SCA-off` 里 `ctrl_shift = 0.000000`
- `SCA-on` 里 `ctrl_shift` 从 `0.0569` 逐步下降到 `0.0348`，但始终非零
- `gate_mean` 没塌到 0 或 1
- `gate_std` 也不是 0

这说明你的 control 路线是真在工作，不是空概念。

这组结果能说明什么

现在可以说明：

- 工程链路是通的
- evaluator 基本可信
- baseline 正常
- SCA-off / on 的区分正常
- SCA 的核心机制成立
- 中间版趋势支持你的论文 claim

但还不能直接当成最终论文 Table 1，因为你现在仍然用了：

- `max_batches_per_epoch`
- 没有 validation best checkpoint
- 还没有正式日志/metrics 保存闭环
- 还没有完全统一“最佳轮次”报告

所以这组最准确的定位是：

第一阶段已完成“中间版可信趋势验证”。

你现在最该做的下一步

直接进入正式版 **Table 1** 收尾，不要再继续做更多中间版了。

下一步该做的不是改模型，而是补实验规范：

1. 加 validation 评估与 best checkpoint 选择
2. 保存 `log.txt` 和 `metrics.json`
3. 去掉 `max_batches_per_epoch`，或者至少先提高到接近全量
4. 在同一规范下正式跑：
    - LightGCN
    - SCA-off
    - SCA-on

你现在的阶段性结论，可以先这样记

你可以先在自己的实验记录里写：

> 在 ML-1M 的中间版实验中，SCA-on 在 Recall@10、NDCG@10 和 HR@10 上均略优于 LightGCN，并明显优于 SCA-off；同时，SCA-on 的 `ctrl_shift` 持续为非零，而 SCA-off 的 `ctrl_shift` 恒为 0，说明 semantic control 的确在决策层发挥作用。

这句话现在是站得住的。

我给你的直接判断

可以推进正式版。
不用再怀疑方向了。你现在缺的已经不是“方法对不对”，而是“把当前趋势固化成可投稿表格”。

下一条我建议我直接给你：
正式版第一阶段需要改的 **`run.py` / logger / best checkpoint** 方案。